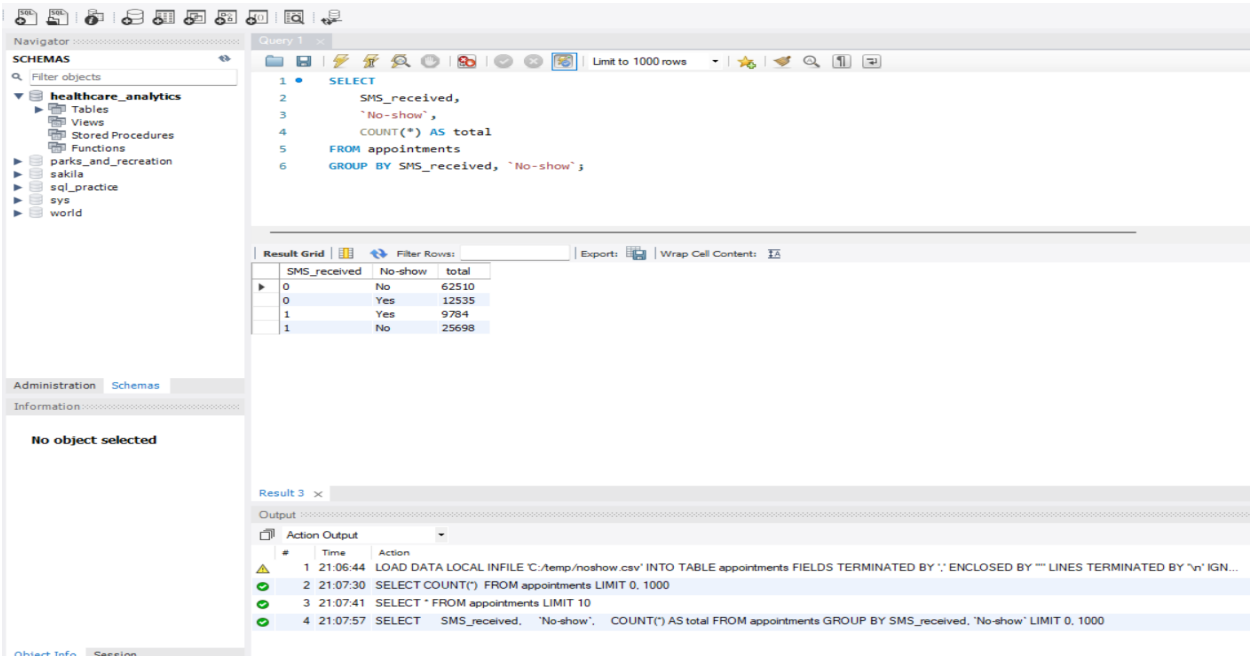
# Healthcare Appointment No-Show Analysis

## Dataset Overview

The dataset was sourced from Kaggle and originates from a publicly available healthcare appointment dataset used for analytics practice. Although the data does not represent a specific hospital system, it contains realistic operational variables such as appointment scheduling dates, patient demographics, reminder indicators, and attendance outcomes. The purpose of this project was to demonstrate analytical techniques using SQL and Power BI in order to identify patterns related to missed healthcare appointments rather than to make clinical conclusions.

## SQL Analysis

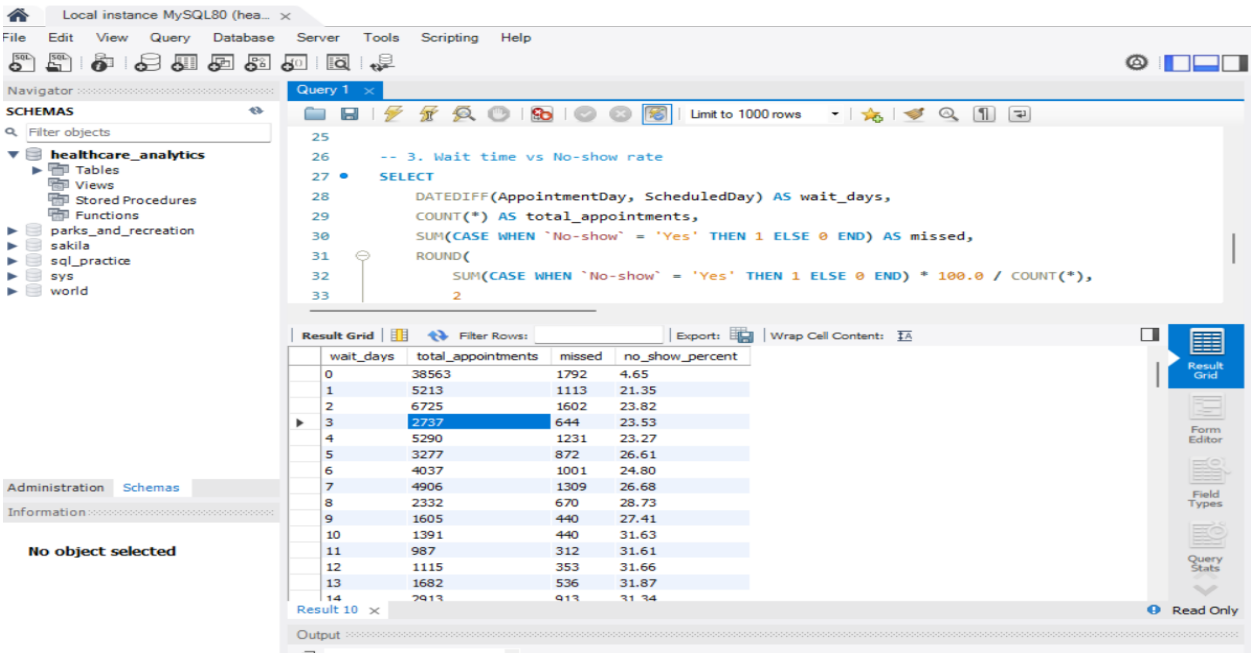
I queried a dataset containing approximately 110,000 medical appointments to investigate whether SMS reminders reduced missed visits. The query grouped appointments by reminder status and attendance outcome. Patients who received reminders had a 27.6% no-show rate, while those without reminders had a 16.7% no-show rate. Initially, I expected reminder messages to reduce the number of missed appointments. However, the results were surprising. Patients who received SMS reminders actually had a higher observed no-show rate compared to patients who did not receive reminders.




This unexpected result suggested that reminder messages may have been associated with higher-risk appointments rather than directly reducing missed visits. For example, reminders may be used more frequently for appointments scheduled further in advance or for patients with a history of missed visits.

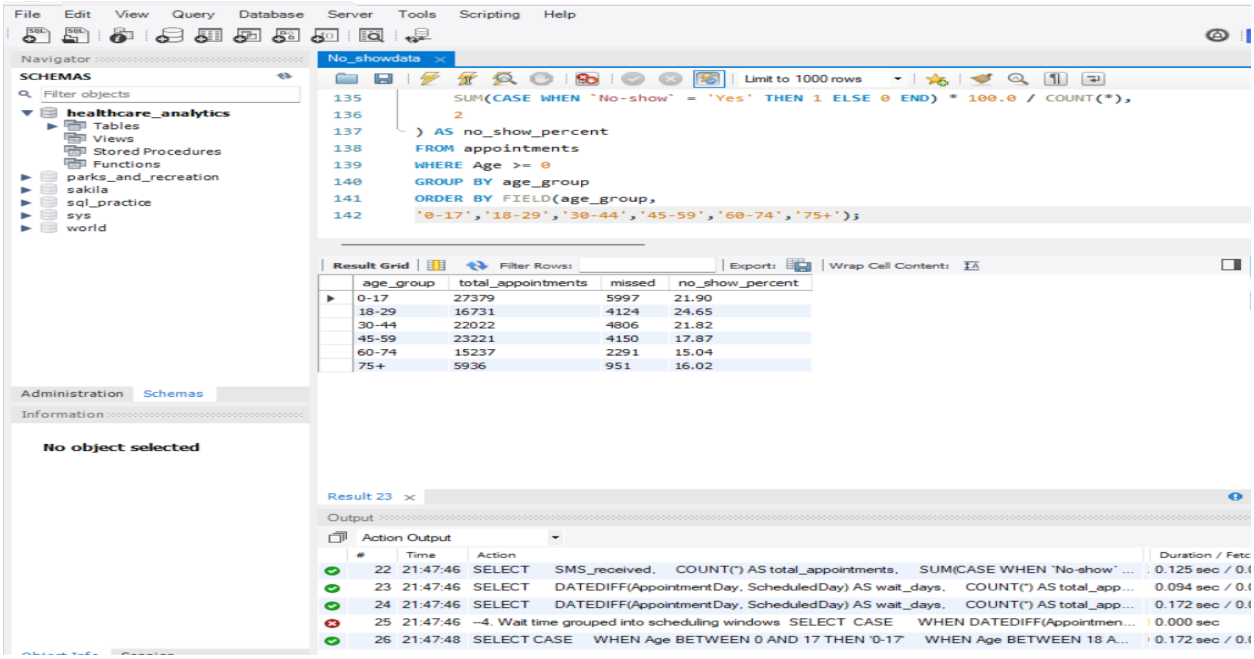
## Scheduling Delay Analysis

To investigate additional factors influencing attendance, I analyzed the relationship between scheduling delay and missed appointments. A strong pattern emerged. Same-day appointments had a no-show rate of approximately 4–5%, while appointments scheduled more than 10 days in advance had no-show rates above 30%. This indicates that longer scheduling delays significantly increase the likelihood that patients will miss their appointments.




## Patient Demographics

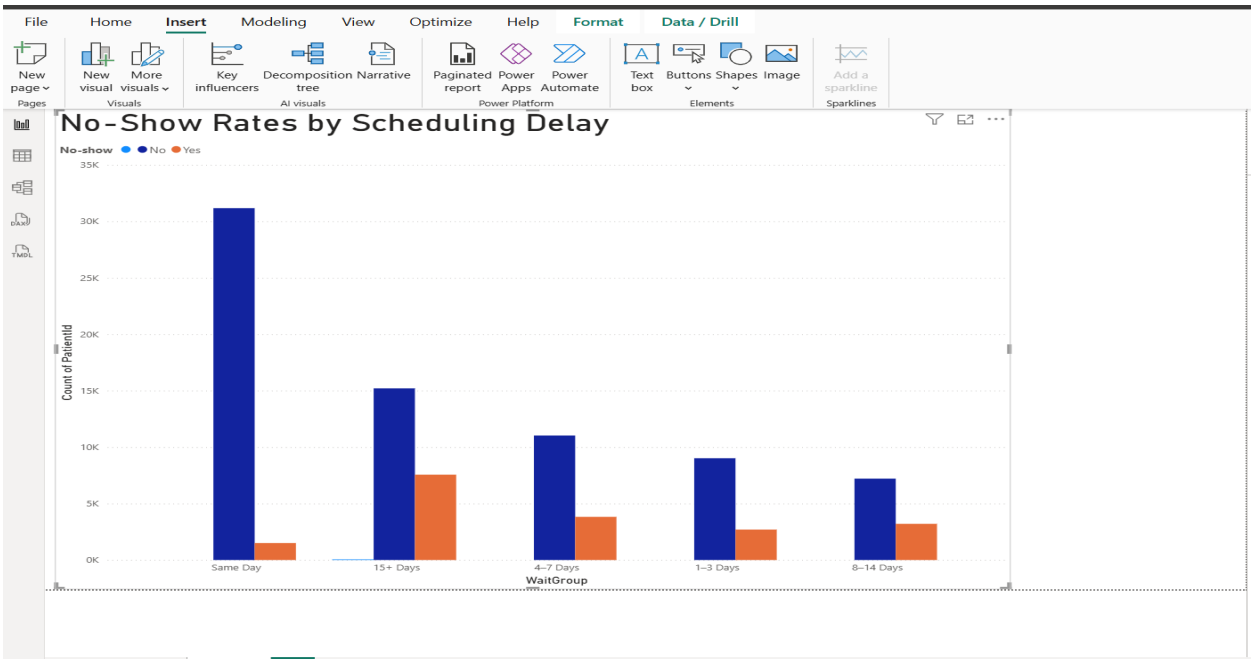
Patient age also appeared to influence attendance behavior. Younger patients, particularly those between 19 and 35, had the highest no-show rates, while older patients were more likely to attend their appointments consistently.



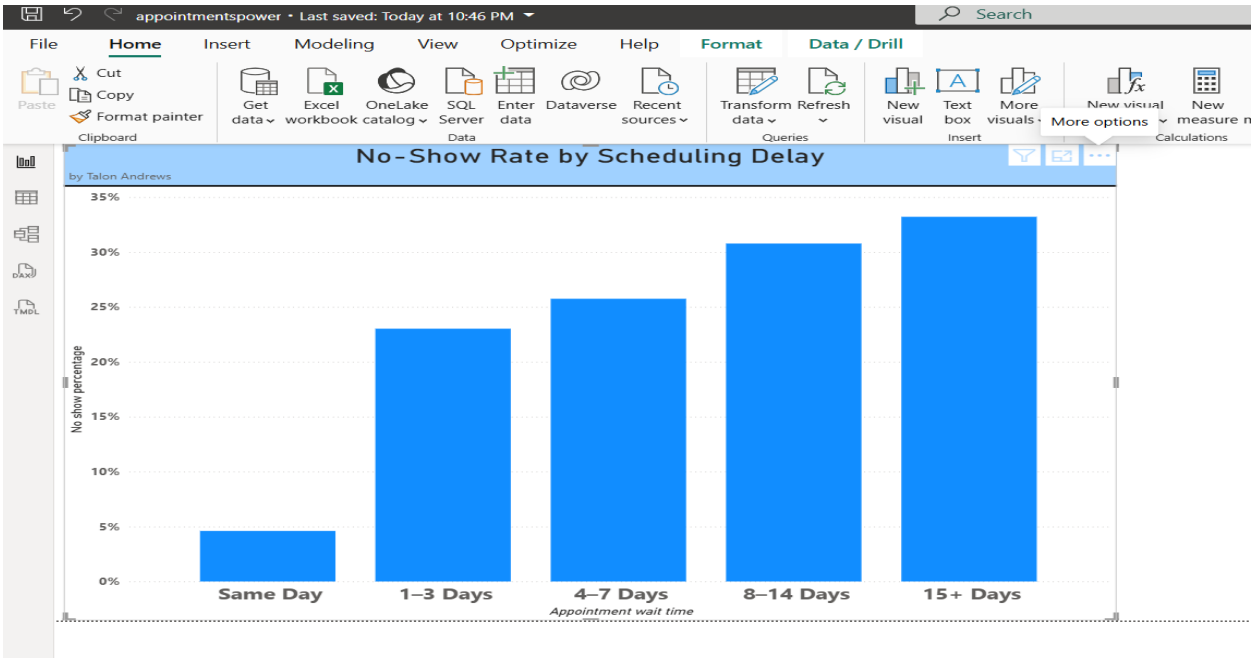

Additional variables such as gender, neighborhood, financial assistance status, and day of the week were also explored. However, these variables did not show strong or consistent relationships with missed appointments in this dataset.

## Power BI Visualization

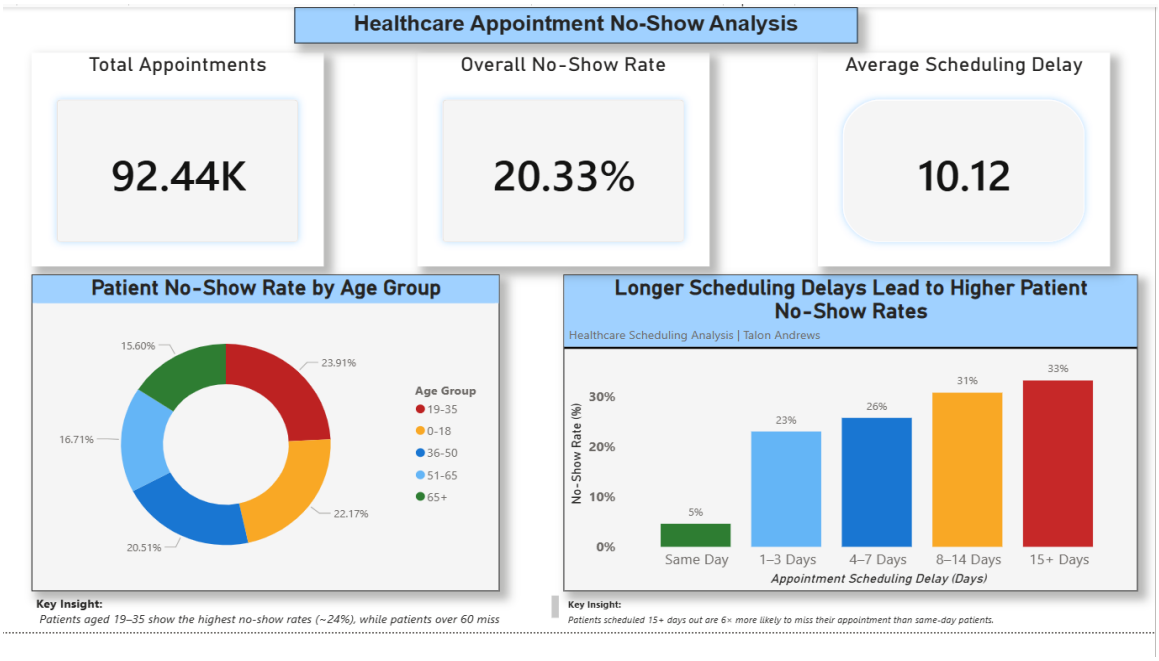
After completing the SQL analysis, the results were visualized in Power BI to make patterns easier to interpret and communicate. The first version of the visualization displayed both “No-show: No” and “No-show: Yes,” which created a double negative and made the chart harder to interpret quickly.



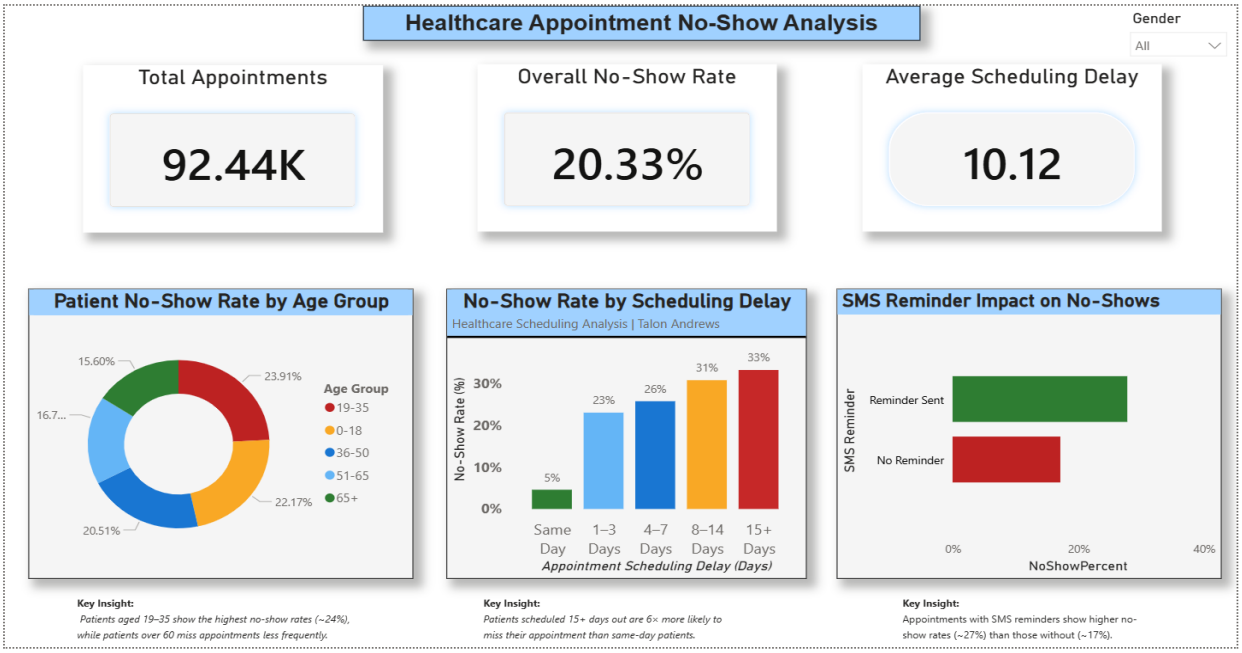

To improve clarity, the chart was redesigned to focus directly on the percentage of missed appointments by scheduling delay category. This made the trend much easier to understand and communicate.




During dashboard development, multiple visualizations were combined to better understand how operational factors such as scheduling delay and patient demographics interact with appointment attendance.




The final dashboard integrates key performance indicators, demographic analysis, and scheduling delay insights into a single visual report that allows viewers to quickly understand the main drivers of missed appointments.




## Project Summary

Using SQL and Power BI, this project analyzed more than 110,000 outpatient medical appointments to identify factors associated with missed visits. The analysis revealed that scheduling delay was the strongest predictor of missed appointments. Same-day visits had very low no-show rates, while appointments scheduled more than 10 days in advance had no-show rates around 30%. Younger patients were also more likely to miss appointments than older patients. These findings demonstrate how operational healthcare data can be analyzed to uncover patterns that may help healthcare organizations better understand and potentially reduce missed appointments.